

Adam Mrotek

Senior Software Engineer

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SUMMARY

Senior software engineer with 4+ years' commercial experience leading front-end delivery on complex web applications and owning them end-to-end - from TypeScript design systems and type-safe patterns to hands-on Python / FastAPI backends. Currently building a production AI product on top of LLMs.

HIGHLIGHTS

- **Croud** — Co-architected a TypeScript design system and led front-end delivery across a 3,000-user platform; built core front-end for an in-house build that replaced a £17M agency quote.
- **Hullabalook (B2B SaaS)** — Shipped client-facing apps and prototypes that drove upsells alongside client directors; mentored and onboarded engineers.
- **VillageOS** — Solo-built a live AI product end-to-end, design to infrastructure; multimodal LLM extraction (text + images) with a golden-dataset eval harness.

TECHNICAL SKILLS

- **Languages:** TypeScript • JavaScript • Python
- **Frameworks / Libraries:** React • Next.js • Node.js • TanStack Query • FastAPI • Vue 3
- **AI / LLM:** OpenAI • Claude / MCP • instructor • Pydantic • RAG • evals • PostHog experiments
- **Infrastructure:** AWS (Lambda, SAM) • Supabase • PostgreSQL • Docker • GitHub Actions
- **UI & Design:** shadcn/ui • Tailwind CSS • Storybook • Figma

WORK EXPERIENCE

Full-Stack Engineer, Croud

Jul 2023 - Mar 2026 (2y 9m)

Marketing Agency

React, TypeScript, Next.js, TanStack Query, shadcn/ui, FastAPI, Tailwind CSS, Vue 3, Docker, Claude / MCP

- **Core Platform (Croud Control, 3,000+ users):** Led front-end delivery across a marketing-automation suite; co-architected the TypeScript component design system and drove the migration of legacy surfaces to type-safe patterns, accelerating release velocity across product areas.
- **Social Influencer Platform:** Early engineer on an in-house influencer-marketing platform - an internal build that replaced a £17M third-party agency quote. Built user profiles, dashboards and registration-of-interest flows - against strict Figma specs, integrating with the FastAPI backend. Also set up a Figma-to-code pipeline - a custom Claude sub-agent that converts designs directly into our codebase - driving cross-team adoption.
- **Internal AI products:** Owned roadmap, scoping and prioritisation for internal AI tools featured in client decks - working closely with design and defining the API contract the backend built to.

Customer Engineer, Hullabalook

Apr 2022 - Jul 2023 (1y 4m)

Improving retail experience using data analytics · Retail · B2B · SaaS · eCommerce

React, Vue.js, Vuex, Nuxt, TypeScript, Fabric.js, Python, GCP

- Drove successful client upsells by prototyping new features, and improved Lighthouse / SEO scores to lift app performance across multiple clients.
- Built robust applications within cross-functional product teams via CI/CD, including highly visual interfaces using Fabric.js and the HTML5 Canvas API.
- Improved data accuracy by building programmatic solutions for unstructured, incomplete datasets using Python.
- Mentored junior engineers through code reviews and pairing sessions, helping raise code quality and shorten onboarding.

Architectural Assistant (Part I & II), Bryden Wood

Mar 2018 - Sep 2021 (3y 6m)

Multidisciplinary architectural company · Property · Design · Construction

- Designed and contributed to architectural projects for airports and data centres, applying platform-based, systems-thinking design principles and driving collaboration across architectural and M&E design teams.

FEATURED PRODUCT

VillageOS — Founder & Full-Stack Engineer (Self-employed)

Apr 2026 - Present

Solo-built AI product · live in production (public demo + deployed API) · villageos.co.uk

Next.js, React, TypeScript, TanStack Query, shadcn/ui, FastAPI, AWS Lambda, Supabase, OpenAI, Groq, PostHog

Designed, built and shipped VillageOS to production solo - a live AI product that turns scattered parent communication (WhatsApp threads, PDF newsletters, photos of flyers) into schema-valid calendar events. Owned every layer end-to-end: frontend, backend, data, infrastructure, CI/CD and LLM evaluations.

- **Typed on both sides** - Pydantic v2 is the single source of truth for the LLM extraction boundary (structured output) and FastAPI request/response validation; the TypeScript client mirrors the same shapes, so the contract is enforced at the API edge.
- **Structured AI, not vibes** - instructor + Pydantic force a strict schema on every extraction, across multimodal (text + vision) inputs, with a golden-dataset eval harness tracking per-field accuracy, latency and cost.
- **Real infrastructure** - FastAPI on AWS Lambda (arm64) via SAM, Supabase JWT auth tied to Postgres Row Level Security, released through OIDC GitHub Actions.
- **Online A/B in production** - a server-assigned PostHog experiment on the live extraction model, field-edit-rate metric with latency and cost guardrails - closing the loop from offline eval to real user behaviour.

EDUCATION

Anthropic Academy

May 2026

Certificate of completion - Claude with Amazon Bedrock

- Building production-ready AI applications with Claude on Amazon Bedrock - the Bedrock API, prompt engineering and evaluations, tool use, retrieval-augmented generation (RAG), the Model Context Protocol (MCP), and agents.

MSc & BArch, Architecture

2011 - 2017

Politechnika Poznańska (Poznan University of Technology) · incl. Erasmus, Università di Firenze

ADDITIONAL

Languages: English (Full Professional), Polish (Native / Bilingual)

Interests: Architecture, tech for good / green tech, feng-shui, cooking.